

PR-2

PART - A

Q.1 What is Regularization in Machine learning?
Why is it needed?

→ Regularization adds penalty term to the loss function to reduce model complexity

- It prevent overfitting by shrinking coefficient values.
- Without regularization, the model memorizes training data and fails on new data
- The two main types are L_1 (Lasso) and L_2 (Ridge).

Q.2 Difference between Ridge Regression (L_2) and Lasso Regression (L_1)

Feature	Ridge (L_2)	Lasso (L_1)
Penalty	Sum of squared coefficient	Sum of absolute coeffi
Feature selection	No - keeps all features (shrinks them)	Yes - can zero out features.
Output	Dense solution	Sparse solution
Best for	Multi collinear feature	High dimensional, Sparse problem.

Q.3 What is cross-validation and why is it imp?

→ Cross-validation evaluates model generalization by splitting data into multiple train/validation sets and averaging the performance score.

- Avoids overfitting to one lucky train-test split.
- Gives a reliable estimate of model performance on unseen data.
- Used for hyperparameter tuning (e.g., finding the best alpha in ridge/lasso).

Q.4 Explain the following cross-validation techniques

K-Fold Cross-Validation

→ Data is split into k -equal folds. Model trains on $k-1$ folds and validates on the remaining 1. This is repeated k -times and results are averaged.

Stratified K-Fold Cross-Validation

Same as k -fold but each fold preserves the same proportion of target value. Crucial when the target is skewed or imbalanced.

Leave - One - Out - Cross - Validation (LOOCV)
 every single becomes a validation set once. very thorough but extremely slow for large dataset.

Time series split
 folds always move forward in time - train on past, validate on future. This prevents temporal data leakage.

Q. 5. Why are tree-based model less sensitive to feature scaling?

→ Decision Trees split on feature threshold (e.g., $\text{area_sqft} > 1500$), not on distance or dot product.

- Multiplying or scaling a feature does not change which threshold is best.

- Random Forest and other tree ensembles inherit this property.

- Contrast: Ridge, Lasso, SVR rely on magnitude - scaling is mandatory for them.